

EXHIBIT 47

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

In the Matter of CERTAIN MOBILE ELECTRONIC DEVICES AND COMPONENTS THEREOF	Investigation No. 337-TA-_____
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**COMPLAINT UNDER SECTION 337 OF THE
TARIFF ACT OF 1930, AS AMENDED**

<u>Complainants:</u> trinamiX Sensing LLC 7800 Shoal Creek Blvd Ste 200N TX 78757 Austin, TX 78757, USA trinamiX GmbH Industriestraße 35 67063 Ludwigshafen am Rhein <u>Counsel for Complainants:</u> S. Alex Lasher Carl Spilly QUINN EMANUEL URQUHART & SULLIVAN, LLP 555 13 th St NW Washington, D.C. 20004 Tel.: (202) 538-8000	<u>Proposed Respondent:</u> Apple Inc. One Apple Park Way Cupertino, CA 95014 (408) 996-1010
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**EXPERT DECLARATION OF IOANNIS KAKADIARIS, Ph.D. REGARDING
INFRINGEMENT TESTING OF iPhone 17**

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I. INTRODUCTION

1. My name is Ioannis Kakadiaris, Ph.D., and I have been retained by counsel on behalf of Plaintiffs trinamiX GmbH and trinamiX Sensing LLC in connection with the above-captioned action. Counsel has asked me to perform testing and offer an expert opinion as to whether the Apple iPhone 17 considers material properties when performing Face ID.

2. I am being compensated at my standard rate of \$600 per hour for my working time and \$250 per hour for my travel time, plus reasonable out-of-pocket expenses. I have received no additional compensation for my work in this litigation, and my compensation does not depend upon the contents of this report, the results of my testing, any testimony that I may provide, or the ultimate outcome of this litigation.

3. Based on the testing that I describe in detail below, it is my opinion that the iPhone 17 considers material properties (skin vs. non-skin) when performing Face ID. As my testing shows, Face ID that successfully unlocks the iPhone in the control test then fails when the material of the face changes (*i.e.*, the skin of the face is covered with a different material—for the purpose of this testing, with Healing Earth), even when the geometry stays the same.

II. QUALIFICATIONS

4. Below is an overview of my background and qualifications.

5. I earned a Bachelor of Science in Physics in 1989 from the University of Athens in Athens, Greece. I earned a Master of Science degree in Computer Science in 1991 from Northeastern University. In 1997, I earned a Doctor of Philosophy degree in Computer Science from the University of Pennsylvania.

6. From 1997 to 2003, I worked for the University of Houston (“UH”) as an Assistant Professor in the Department of Computer Science.

7. From 2003 to 2006, I worked as an Associate Professor with tenure in the Department of Computer Science of the University of Houston. From 1997 to 2005, I served as the Director of the Visual Computing Lab in the Department of Computer Science, UH.

8. Since 2005, I have served as the Director of the Computational Biomedicine Lab in the Department of Computer Science, University of Houston.

9. I have extensive experience teaching UH Computer Science course 6373, which covers the principles of computer vision. I am an expert in facial recognition and object recognition, as described in my research publications.

10. Since 2011, I have been a tenured Hugh Roy and Lillie Cranz Cullen Distinguished University Professor at the University of Houston. My research has focused on facial recognition. My team and I have contributed to 3D face (and ear) recognition, 3D-aided 2D Face Recognition, 2D-2D Face Recognition, and profile-based face recognition. CBL's 3D-3D face recognition software ranked first in the 3D-shape section of the 2007 Face Recognition Vendor Test (FRVT) organized by NIST, while CBL's 3D-2D method outperforms the state-of-the-art 2D face recognition methods. Currently, CBL researchers are addressing critical challenges including low resolution data, indoor/outdoor illumination, accurate landmark and pose estimation, cross resolution matching, and score normalization.

11. I have served as an Associate Editor of 1) IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) (premier venue for computer vision work); and 2) Transactions on Biometrics, Behavior, and Identity Science (T-BIOM) (premier venue for face recognition work).

12. I have authored numerous publications in face recognition, including journal articles, book chapters, and conference papers.

13. I have been active in several professional organizations and conferences.

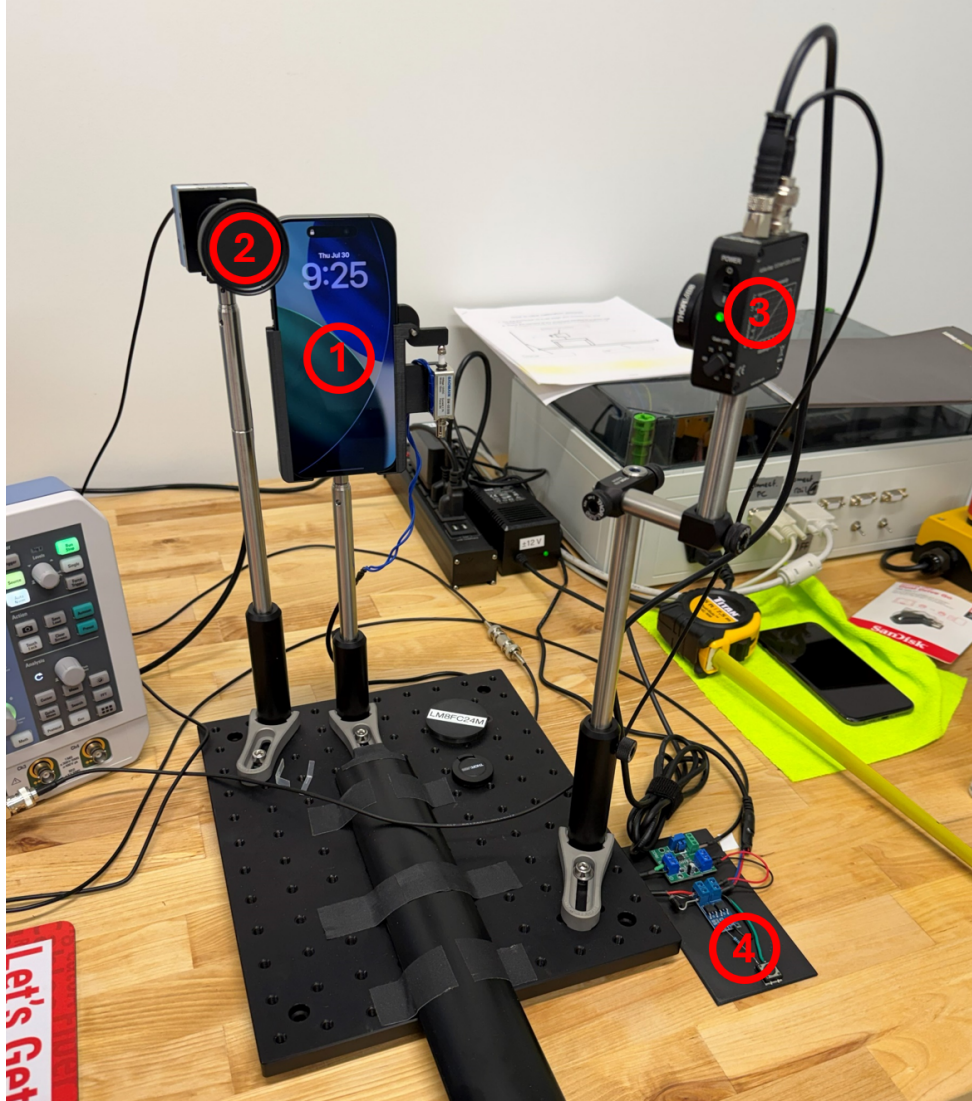
14. I have received several honors and awards. In addition to winning the UH Computer Science Research Excellence Award twice, I have been recognized with several distinguished honors, including the NSF Early Career Development Award, the Schlumberger Technical Foundation Award, the UH Enron Teaching Excellence Award, and the James Muller Vulnerable Plaque Young Investigator Prize. My face recognition research has been featured on the Discovery Channel, National Public Radio, KPRC NBC News, KTRH ABC News, and KHOU CBS News.

15. Additional details of my education and employment history, professional service, patents, publications, and other testimony are set forth in my current curriculum vitae.

III. TESTING

A. Protocols and Procedures

16. I used the following configuration to test Face ID unlocks:



17. As shown in the image above, the configuration includes (1) an Apple iPhone 17 equipped with Face ID¹; (2) an near infrared (“NIR”) global shutter camera (The Imaging Source DMM 37UX265-ML) to take images of the object in front of the front-facing camera of the phone during testing, including any laser illuminations output from the phone; and (3) a photodiode (Thorlabs PDA36A2) and (4) controller (Picoscope 2204A) that trigger the camera during testing

¹ The iPhone used in this testing was an Apple iPhone 17, model number MG464LL/A, having serial number FDJLFC6L26 and operating on iOS version 26.5.2.

in response to illumination from the phone. For each unlock attempt, this configuration records a series of images that are later inspected and analyzed.

18. To confirm the test results received with a first test subject, the tests were reproduced with a second test subject. Each test subject performed unlock attempts with either a clean face or a face covered entirely with *Luvos Heilerde 2 hautfein Akne Pulver* (“Healing Earth”),² which is an acne powder that, for the purpose of this testing, was used to alter the material properties of the test subjects’ skin without altering the geometry of the face. For the testing involving Healing Earth, two parts of the Healing Earth powder was mixed with one part water until it became a homogeneous paste for testing that is opaque enough to alter the skin’s reflection characteristics, yet thin enough to be applied without altering the geometry of the face. The test subjects had their faces sprayed with water right before testing to prevent the Healing Earth from drying during testing.

19. The following images show how each test subject appeared in the control test (with a clean face) and in the test with Healing Earth applied:

² <https://www.luvos.de/anwendungsgebiete/akne-und-unreine-haut/>

Test Subject #1:



Test Subject #2:



20. The following procedures were followed during testing: Before starting any unlock attempts, the test subjects enrolled their faces in Face ID on the iPhone using a clean face. Then, I performed a series of 10 unlock attempts on each test subject with a clean face, using the configuration shown above. For each unlock attempt, the subject was placed in a central, frontal pose relative to the iPhone front camera, approximately 40 centimeters away, measured from below the nose to the upper front edge of the iPhone. At each unlock attempt, I documented the result, and the configuration above captured a series of images stored in a manner attributable to each individual unlock attempt.

21. Once the control testing concluded, the test subjects covered their entire faces with Healing Earth, and I performed a series of 10 unlock attempts on each subject using the same parameters described above (*e.g.*, placing each test subject in a central and frontal pose with respect to the iPhone front camera, approximately 40 centimeters away from the camera, measured from below the nose to the upper front edge of the iPhone).

22. I understand that after three unsuccessful unlocks, the iPhone requires manual unlocking by entering the PIN code. I understand that if a manual unlock occurs shortly after Face ID fails for the third consecutive time, the iPhone may update its internalized face model to continuously learn any changes in the user's face. Accordingly, whenever Face ID failed three consecutive times and required a manual unlock, I removed the test subject's face from view of the phone before manually entering the PIN code in order to avoid altering the enrolled face data. I then had the test subject resume the testing position described above and continue with the unlock attempts. At each unlock attempt, I documented the result, and the configuration above captured a series of images stored in a manner attributable to each individual attempt.

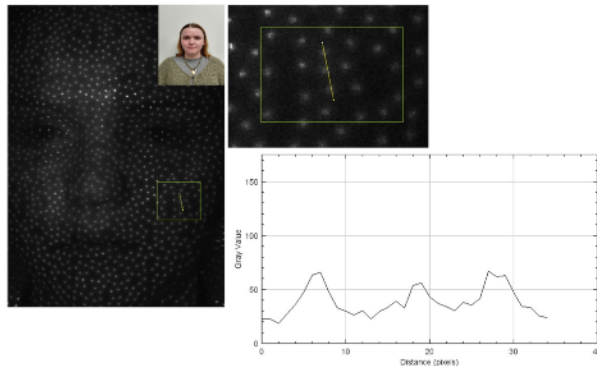
B. Analysis and Results

23. Once testing was completed, the resulting images were analyzed both visually and using ImageJ to determine whether the different materials captured by the iPhone (skin and Healing Earth) produced different reflection characteristics.

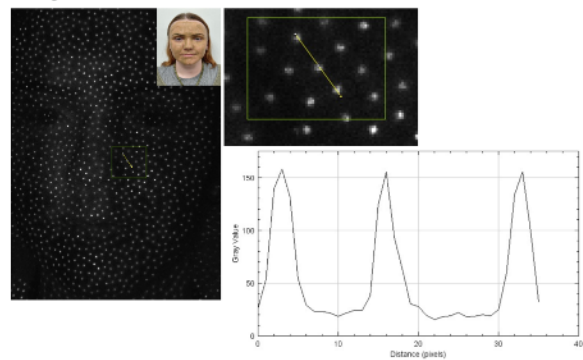
24. In the images taken when skin was showing (an example is provided below), a combination of diffuse reflection and volume scattering caused the laser illumination spots output by the iPhone to broaden, resulting in a brighter background and lower peak intensity. In the images taken when the skin was covered with Healing Earth, by contrast (an example is provided below), Healing Earth acted as a diffuse, reflective material only, resulting in narrower illumination spots, darker illumination between the spots, and higher peak intensity of the illumination spots.

25. These qualitative results were reflected in the quantitative results produced using ImageJ:

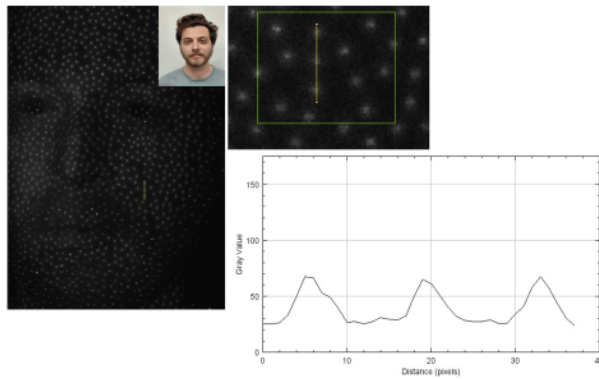
Subject01 Covered area: none – natural Face



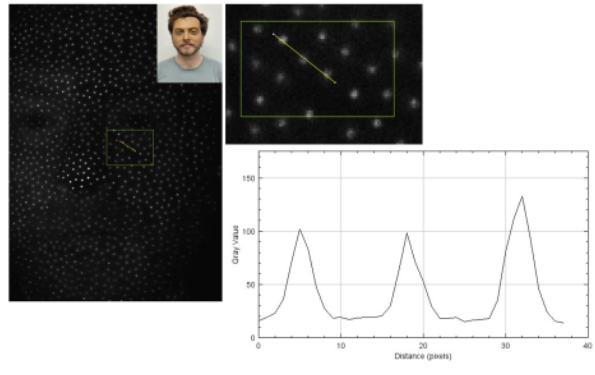
Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face



26. On the left are the test subjects with clean faces, and on the right are the test subjects with faces covered with Healing Earth. The analysis focuses on three dots projected on the test subjects' faces.

27. Each dot spans multiple sensor pixels, and the x-axis shows distance in pixels. The grey value for each pixel is shown on the y-axis.

28. The graphs show the distribution of grey-value intensity in the cross-section of each dot. The distribution shows a low grey value at the edge of each dot and the maximum grey value toward the center.

29. The graphs further show the differences in the reflection characteristics of the dots depending on whether Healing Earth is applied to the skin. For skin, the maximum intensities of the dots are comparably low, and the peaks are broadened. For Healing Earth, the maximum intensities are high, the peaks are sharp, and the background or base signal is comparably low.

30. Finally, the results of the unlock test are shown below:

Test Administered	Test Subject	Results	
		Unlock	No Unlock
Without Healing Earth	Test Subject #1	10	0
	Test Subject #2	10	0
With Healing Earth	Test Subject #1	0	10
	Test Subject #2	0	10

31. In my opinion, these results show that the iPhone 17 considers material properties, and thereby distinguishes skin from non-skin when performing Face ID. In other words, the performance of Face ID on the iPhone 17 changes when the material of the face (*e.g.*, skin or Healing Earth) changes, even when the geometry of the face stays the same (*i.e.*, when the same test subject is presented to Face ID). Accordingly, these test results show that the iPhone 17 considers material (skin vs. non-skin) when performing Face ID.

IV. CONCLUSION

32. For the reasons discussed above, it is my opinion that the iPhone 17 considers material properties, and thereby distinguishes skin from non-skin, when performing Face ID.

* * *

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true. I further declare that these statements were made with the knowledge that willful false statements and the like are punishable by fine or imprisonment, or both, under 18 U.S.C. § 1001.

Date: August 26, 2026

loannis
Kakadiaris

Digitally signed by
Ioannis Kakadiaris
Date: 2026.08.26
16:55:03 -05'00'

Dr. Ioannis Kakadiaris